

Can Spectral Information Work While Extracting Spatial Distribution?—An Online Spectral Information Compensation Network for HSI Classification

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Abstract—In the past few years, deep learning-based methods have shown commendable performance for hyperspectral image (HSI) classification. Many works focus on designing independent spectral and spatial branches and then fusing the output features from two branches for category prediction. In this way, the correlation that exists between spectral and spatial information is not completely explored, and spectral information extracted from one branch is always not sufficient. Some studies also try to directly extract spectral-spatial features using 3D convolutions but are accompanied by the severe over-smoothing phenomenon and poor representation ability of spectral signatures. Unlike the above-mentioned approaches, in this paper, we propose a novel online spectral information compensation network (OSICN) for HSI classification, which consists of a candidate spectral vector mechanism, progressive filling process, and multi-branch network. To the best of our knowledge, this paper is the first to online supplement spectral information into the network when spatial features are extracted. The proposed OSICN makes the spectral information participate in network learning in advance to guide spatial information extraction, which truly processes spectral and spatial features in HSI as a whole. Accordingly, OSICN is more reasonable and more effective for complex HSI data. Experimental results on three benchmark datasets demonstrate that the proposed approach has more outstanding classification performance compared with the state-of-the-art methods, even with a limited number of training samples.

Index Terms—HSI classification, deep learning, online spectral information compensation, progressive filling, multi-scale features.

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I. INTRODUCTION

HYPERSPECTRAL image (HSI) has hundreds of narrow adjacent bands, which are rich in spectral signatures and strongly correlated with spatial features. Owing to exceptional properties, HSI is able to distinguish a variety of complex objects, which is a difficult task for other remote sensing images [1]. The process to achieve the above task is HSI classification. HSI classification aims to classify each pixel into a given label, such as water, grassland, or building [2]. It has been applied to many scenarios with great success, such as urban development [3], [4], environmental monitoring [5], and precision agriculture [6], [7].

However, different from the general classification problem [8], [9], HSI classification is more challenging on account of the curse of dimensionality [10], [11], [12] and the resulting higher computational costs [13], [14], [15]. To improve classification performance, feature selection and feature extraction methods are employed at first. The goal of feature selection is to retain the channels that are most representative and useful for classification and abandon those that do not contribute much [16], [17]. Many feature selection methods [18], [19] are proposed to reduce band redundancy. Nevertheless, the above-mentioned approaches use the same spectral bands for all pixels in different categories, ignoring their discrepancies in spectral sensitivity. Since optimal spectral bands for classification are diverse for each category, a distinctive combination of spectral bands is needed to apply to different categories for classification [20], [21], [22]. As a result, feature extraction methods are proposed to combine these spectral bands together via some linear or nonlinear transformations. Typical feature extraction methods include principal component analysis (PCA) [23], independent component analysis (ICA) [24], linear discriminant analysis (LDA) [25], and kernel-based strategies [26]. The above traditional methods rely heavily on the prior knowledge of experts and require manual feature design, which is so tedious and time-consuming. Once the designed features are not appropriate, a large amount of information will be lost. Therefore, these methods are poorly adapted and generalized to different or invisible datasets. In addition, these handcrafted techniques are always designed as “shallow and low-level” models, implying

that their capacity to process complex nonlinear information is restricted.

Over the past several years, deep learning-based methods with hierarchical feature learning have attracted significant attention [27], [28]. As “deep and high-level” models, they can automatically extract more valuable features and have shown powerful superiority in HSI classification [29]. In [30], Hu et al. employ a convolutional neural network (CNN) to capture spectral signatures. In order to take multi-scale spatial information into account, Khotimah et al. [31] propose a two-stream spectral-spatial residual network (TSRN). Nevertheless, spectral information extraction and spatial information extraction are completely separated in TSRN, resulting in the lack of HSI integrity. Moreover, Zhong et al. [32] use 3D convolutional layers to directly obtain spectral-spatial information of HSI. It is worth noting that 3D convolution does not discriminate between spectral and spatial features, and just mixes all the features in a single stream. Thus, spectral features extracted are usually insufficient. Besides, there is serious over-smoothing phenomenon in the classification maps of 3D CNN-based methods. Recently, several studies [33], [34] have tried to consider HSIs as sequence data. As a result, recurrent neural network (RNN)-related methods are developed for hyperspectral data analysis. Hu et al. [35] use long short-term memory (LSTM) network to realize spectral and spatial joint feature extraction. Pan et al. [36] propose a single gate recurrent unit (GRU) to learn spectral correlation within a whole spectrum input. RNN-based methods have the ability to perform better spectral feature representation, but these methods have difficulty in mining spatial information.

Therefore, although many deep learning-based methods have emerged, there are still many problems, which can be divided into three main topics. The first issue is the inequality between spectral feature extraction and spatial feature extraction. The current approaches rely too much on deep learning models of computer vision tasks, with spatial information mining as the main focus, and spectral information is greatly neglected. Obviously, this is inconsistent with the character of rich spectral features in HSI, and fails to make full use of the excellent nature of hyperspectral remote sensing. The second problem exists in the spectral-spatial feature extraction process. Generally, there are two types of existing methods for joint spectral-spatial extraction. One is to design two completely separate branch networks for spectral and spatial feature extraction, respectively. That means spectral domain and spatial domain of HSI are totally independent. Apparently, these kinds of methods fragment the spectral and spatial information into two uncorrelated parts, which violates the integral nature and data structure of HSI cube. To better match HSI cube structure, 3D CNN-based approaches are advanced recently as another way. However, since 3D convolution indiscriminately captures spectral and spatial features, resulting in weaker spectral information extraction. The rich spectral information of HSI is not fully utilized, and the features learned by the network are much coarser and more over-smoothing. Last but not least, the lack of multi-scale spatial information raises the third challenge in HSI classification task. The deeper the neural network is, the richer and more

abstract the extracted features are. Accordingly, a considerable number of approaches are devoted to deepening the network to obtain detailed features. However, generic global features are ignored during this process, leading to the absence of multi-scale features. That means features at different scales, such as large-scale grass and small-scale road, are difficult to simultaneously identify.

Considering the above limitations and challenges of existing HSI classification methods, we propose an online spectral information compensation network (OSICN). The framework of the proposed methodology is shown in Fig. 1, where the detailed architecture of the multi-branch network (MBN) of Fig. 1 is shown in Fig. 2. First of all, a group of candidate spectral vectors are generated via candidate spectral vector mechanism with dissimilar LSTM networks. Then, from the holistic nature of HSI, we propose a progressive filling process instead of splitting it into irrelevant spectral and spatial parts.

Finally, we design MBN to extract multi-scale spatial information. Moreover, the spectral auxiliary loss function, spatial auxiliary loss function, and joint loss function are designed for optimization. The major contributions of this paper can be summarized as follows:

- 1) The candidate spectral vector mechanism is proposed to overcome the inequality problem that exists between spectral information extraction and spatial information extraction. In the spectral domain, spectral signatures are first individually extracted by diverse LSTM networks to produce candidate spectral vectors, and then they are sent to the progressive filling process.
- 2) In order to fully integrate the spectral information into the spatial domain in advance, we propose a progressive filling process with two stages. In this manner, spectral and spatial domains are no longer simply divided, and HSI data is seen as a unified whole, thus enabling the accomplishment of simultaneous extraction and full fusion of spectral-spatial features.
- 3) We propose MBN to capture fine spatial distribution for objects of different scales. MBN is designed with multiple branches and a multi-level extraction process, which can achieve precise extraction of multi-scale features.
- 4) In this study, we find an interesting phenomenon that too much spectral information being sent to the bottom layer will cause interference with the learning of spatial distribution. It is very important to explore the suitable extraction mode and fusion process, and the proposed method provides an idea for online supplementing and combining spectral signatures for HSI classification.

The remainder of this paper is organized as follows. Section II describes the proposed OSICN in detail. Section III reports experimental results and analysis. Conclusions are summarized in Section IV.

II. ONLINE SPECTRAL INFORMATION COMPENSATION NETWORK

Online spectral information compensation network (OSICN) contains the following components: candidate spectral vector

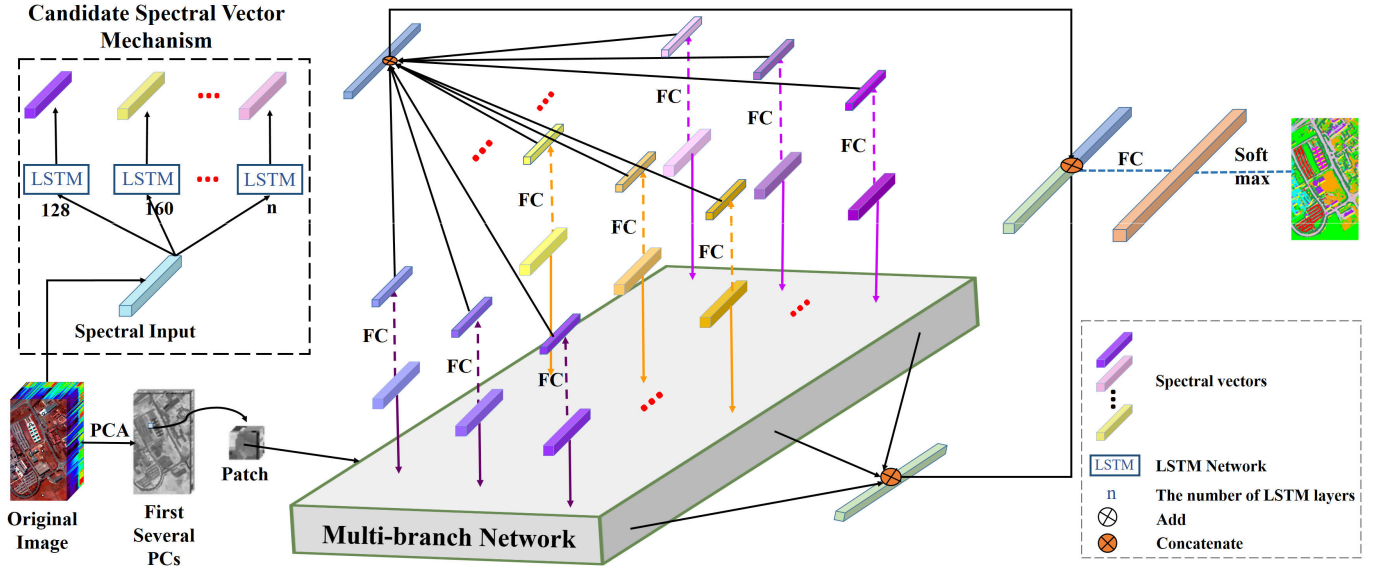


Fig. 1. Flowchart of the proposed OSICN.

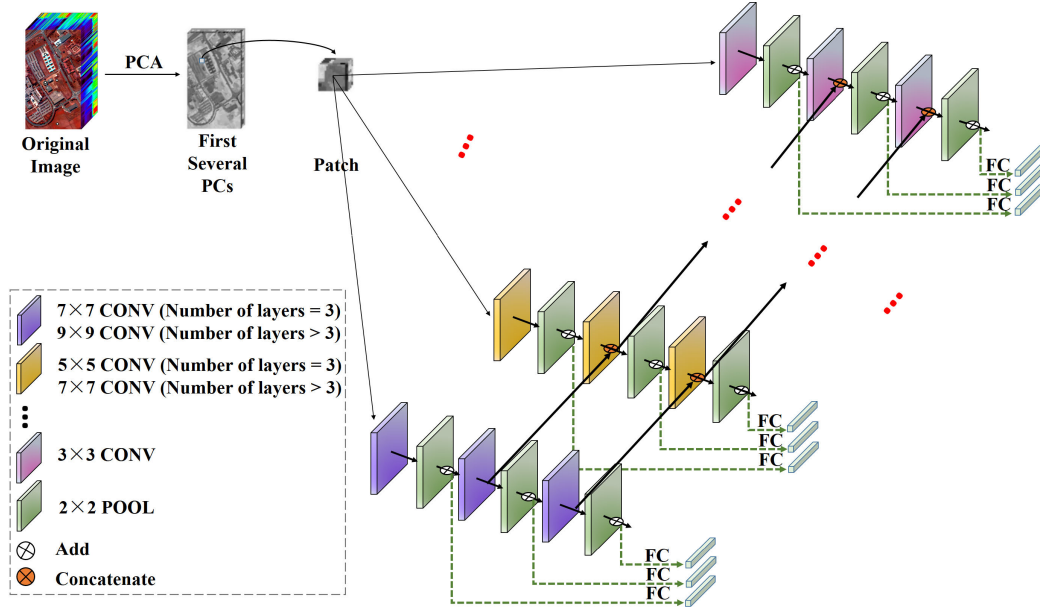


Fig. 2. Architecture of the proposed MBN.

mechanism, progressive filling process, and multi-branch network (MBN). In the following sections, we will present the network structure and optimization of the proposed method.

A. Candidate Spectral Vector Mechanism

Most existing methods are devoted to obtaining the spectral and spatial information separately, which makes the close connection between the spectral and spatial domains broken [37]. In addition, spectral and spatial features are only fused in the last stage, resulting in incomplete integration, and wasting a large amount of potentially valuable information in HSI [38]. Therefore, we propose a new concept: candidate spectral vector mechanism, to distinguishably match them to different levels of spatial features.

To be specific, we employ multiple LSTM networks with different hidden units to generate candidate spectral vectors. Let n denotes the number of hidden units in the k th LSTM network, and it can be defined as follows:

$$n = 128 + 32 \times (k - 1) \quad (1)$$

In this way, a group of candidate spectral vectors $\{s_1, s_2, \dots, s_k, \dots, s_q\}$ ($1 \leq k \leq q$) from q corresponding LSTM networks is generated for spectral information compensation. From Equation (1), it can be concluded that the number of hidden units is proportional to the ordinal number of LSTM networks. The larger the ordinal number value k of LSTM network, the more hidden units there are. As a result, the spectral features extracted are more abstract and robust. Then,

well-designed candidate spectral vectors via the above approach need to be correspondingly matched to different branches of MBN in advance. That is to say, the first stage of initial spectral information compensation is performed during spatial feature extraction. However, how to properly match a group of candidate spectral vectors with the proposed MBN is a key issue that requires careful consideration. As shown in Fig. 2, MBN extracts features at disparate scales with convolutional kernels of various sizes, for example, 7×7 convolutional kernels are used to extract large scale land-covers such as grass and water, while 3×3 convolutional kernels are used to extract small scale objects such as building edges and roads. As targets become smaller and finer, more spectral signatures need to be compensated for fine classification. To this end, deep LSTM networks with many hidden units generate rich candidate spectral vectors to match locally spatial details extracted by the small-scale convolutional kernel. On the contrary, shallow LSTM networks with few hidden units produce limited candidate spectral vectors to match non-locally spatial distribution captured by the large-scale convolutional kernel. Based on the above analysis, spectral signatures can be differentially compensated into the corresponding branch of MBN to achieve the adequate fusion of spectral-spatial features.

B. Progressive Filling Process

To obtain finer spectral-spatial features, and to make full use of the information from HSI, a spectral signature progressive filling process is designed in this section. As displayed in Fig. 3, the whole process can be divided into two stages: initial fusion in the first stage and enhanced fusion in the second stage.

In the first stage, spectral information is engaged in classification in advance and helps spatial features to establish a better learning mechanism, which can optimize classification results. Let $i (1 \leq i \leq l)$ and $j (j = 2, 4, 6)$ represent i th branch and j th layer (the output of the pooling layer) of the proposed MBN, respectively. l is the number of branches in MBN. The spectral information compensation process in the first stage can be expressed as:

$$x_{i',j'} = x_{i,j} + s_k \quad (2)$$

where $x_{i,j}$ is the feature map of the j th layer in the i th branch of MBN, s_k is the candidate spectral vector, and k, i, j gradually increases. $x_{i',j'}$ is the output feature map after the first stage of spectral information compensation of the j th layer in the i th branch, and $+$ represents the addition operation. There are two reasons for using addition scheme. On the one hand, the addition scheme allows for the overlay of spectral and spatial information, thus increasing the amount of information describing HSI. On the other hand, adding operation enables spectral features to be incorporated into spatial features, making the new features generated under the guidance of the spectral features. New feature maps after compensating for the spectral information continue to be fed into the next level of convolution, at which point MBN can perform subsequent feature extraction guided by spectral

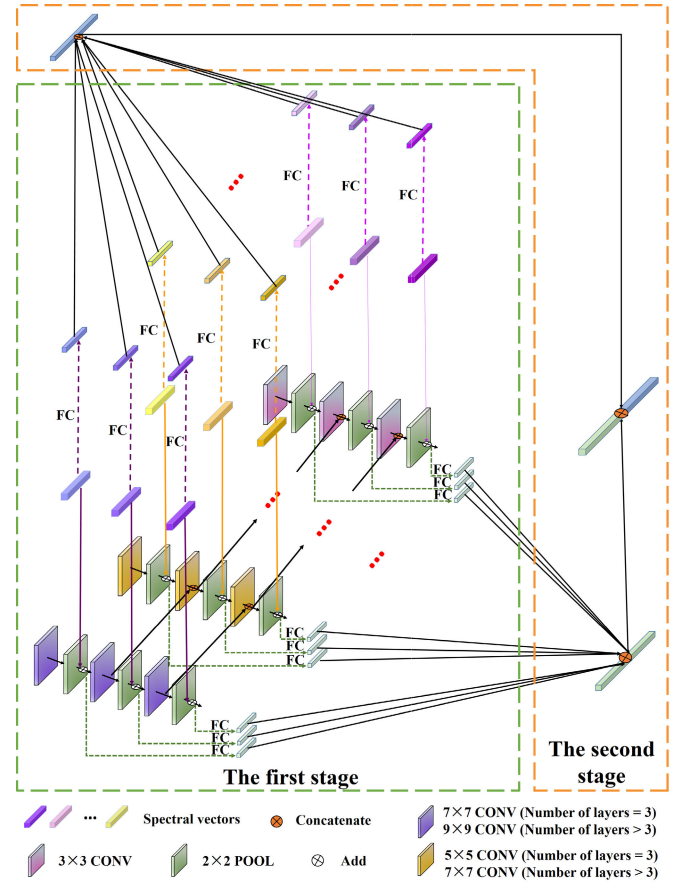


Fig. 3. Progressive filling process.

information. In this way, spatial information is combined with spectral information much earlier and will bring spectral features to participate in learning multi-scale features, thus enabling the network to learn richer and more valuable features just in the first stage of spectral information compensation.

The second stage of the spectral information compensation process aims to intensify the fusion of spectral-spatial information. Specifically, candidate spectral vectors involved in the first stage of spectral information compensation and the features extracted from MBN are fed into the fully connected layer. In the next place, these features are concatenated separately, and finally the results after two sets of concatenation operation are fused, thus complementing spectral information again. In particular, spectral features and spatial features fused in the previous stage are independently sent to fully connected layer and concatenation layer. Afterward, they are fused equally, so spectral information is further and more fully compensated in this stage. The spectral compensation process in the second stage can be defined as follows:

$$Spe = f_{FC}(s_1) \oplus f_{FC}(s_2) \oplus \dots \oplus f_{FC}(s_{q-1}) \oplus f_{FC}(s_q) \quad (3)$$

$$Spa = f_{FC}(x_{1',1'}) \oplus f_{FC}(x_{1',2'}) \oplus \dots \oplus f_{FC}(x_{(l-1)',2'}) \oplus f_{FC}(x_{l',3'}) \quad (4)$$

$$Joint = Spa \oplus Spe \quad (5)$$

where f_{FC} is the function of a fully connected layer, q indicates the number of LSTM networks, and l is the number of

branches of MBN. *Spe* is the result after concatenation of q candidate spectral vectors. Similarly, *Spa* is the result of fusing spectral information and then passing through the concatenation layer. *Joint* is the result of the second stage of spectral information compensation. Especially, $\oplus$ represents concatenation operation, which is different from the first stage. The concatenation scheme splices all the original features so that they can be sufficiently utilized during the subsequent network learning. Furthermore, it allows the network to automatically learn how to integrate features. Consequently, the concatenation operation is more effective to strengthen the union of spectral and spatial features. By multiple concatenation operations in the second stage, spectral features are further fused and compensated, which is useful for accurate category determination.

To sum up, spectral information compensation method consists of a first stage of initial fusion and a second stage of enhanced fusion. The proposed progressive filling process can effectively mine spectral signatures and make them contribute to spatial feature extraction by enhancing the fusion of the spectral-spatial information. To be more specific, in the first stage, since spectral features are added to the network learning while extracting spatial features, information captured is conducted in advance, avoiding the lag and neglect of spectral information. In addition, feature extraction process at each level is compensated with the corresponding candidate spectral vectors, so beneficial information is continuously increased and passed in the network. Therefore, the first stage of filling aids final classification and makes the classification decision more accurate. In the second stage, the spectral vectors involved in the first stage and the features extracted of MBN after compensation are first fused separately, and then further combined together. Hence, in the second stage, not only spectral feature learning is enhanced, but also the compensation process of the first stage is further supplemented.

C. Multi-Branch Network

Convolution kernels with diverse sizes can directly extract multi-scale information [39]. Besides, as the network deepens, features maps extracted are able to provide more abstract and robust information. Therefore, we propose a multi-branch network (MBN) that uses different convolutional kernel sizes at distinct layers to acquire multi-scale information. Also, different spectral information is compensated at corresponding depths to improve the learning performance of subsequent networks. In such a situation, the features flowing through the network are always more distinguished, precise, and useful. In this section, we will outline the structure of the proposed MBN.

As can be seen in Fig. 2, to directly capture multi-scale features from input HSI data, we design multiple branches in parallel with distinct kernel sizes, such as 7×7 , 5×5 , and 3×3 . Branches with larger convolution kernel sizes can extract large-scale spatial information, which is suitable for wide objects such as grassland and water bodies; correspondingly, branches with smaller ones may extract small-scale detailed information, which is appropriate for narrow

objects such as roads and edges of houses. With a multi-branch structure, MBN can directly acquire precise features at various scales. Moreover, for finer local features, features extracted from larger-size convolutional kernels are fed to the branches of smaller-size convolutional kernels. For this reason, the learning of small-scale features can be guided by large-scale information so as to obtain elaborate feature maps with multi-scale feature fusion.

HSI has a large number of bands and high-dimensional property. When the model extracts high-dimensional features of HSI, the number of training samples for backpropagation also increases dramatically. However, the annotation of HSI is very challenging due to environmental variations, sensor errors, and spectral uncertainties, resulting in a scarcity of available training samples. As a result, the small number of labeled training samples for HSI can hardly cope with the requirements of parameter optimization, decreasing the representation capability of the model. It can be seen that although the growth of the number of spectral bands in HSI implies more classification information, classification results are still unsatisfied, giving rise to the Hughes phenomenon (the phenomenon that the classification accuracy increases and then decreases as the number of bands involved in the calculation grows during HSI processing). Therefore, PCA algorithm is employed for dimensionality reduction. Let the original HSI data be $\mathbf{x} \in \mathbb{R}^{r \times c \times b}$, where the spatial size of $\mathbf{x}$ is $r \times c$, and the spectral band number is b . The first p principal components of original data cube are reserved, and the dimension-reduced data is indicated as $\mathbf{x}_p \in \mathbb{R}^{r \times c \times p}$. Around each pixel of HSI, a neighbor patch $\mathbf{x}_{in}$ is extracted with the size of $w \times w \times p$. Denote the i th branch and the j th layer of the proposed MBN by $i(1 \leq i \leq l)$ and $j(1 \leq j \leq 6)$, respectively. l is the number of branches in MBN. $\mathbf{x}_{in}$ is the input of MBN, $\mathbf{x}_{i,out}$ is the output of the i th branch in MBN, whereas the structure of the first branch differs from that of the other branches. The extraction process of the first branches ($i = 1$) can be derived as follows:

$$\mathbf{x}_{i,j} = \begin{cases} \text{ReLU}(\text{Conv}(\mathbf{x}_{in}; \mathbf{W}_{i,j}, \mathbf{b}_{i,j})), & j = 1 \\ \text{ReLU}(\text{Conv}(\mathbf{x}_{j-1}; \mathbf{W}_{i,j}, \mathbf{b}_{i,j})), & 1 < j \leq 5 \end{cases} \quad (6)$$

$$\mathbf{x}_{i,j+1} = \text{MaxPool}(\mathbf{x}_{i,j}) \quad (7)$$

$$\mathbf{x}_{i,j+1} = \mathbf{x}_{i,j+1} + \mathbf{s}_k \quad (8)$$

$$\begin{cases} j = j + 2 \\ k = k + 1 \end{cases} \text{ return (6), } \quad 0 < j < 5 \quad (9)$$

$$\mathbf{x}_{i,out} = \mathbf{x}_{i,j+1}, \quad j = 5$$

It should be noted that the first layer of each branch is a convolutional layer, which is quite distinct from other convolutional layers. Accordingly, the extraction process is also varied according to different layers. The extraction process of the other branches ($1 < i \leq l$) can be divided into the following two ways:

If $j = 1$:

$$\mathbf{x}_{i,j} = \text{ReLU}(\text{Conv}(\mathbf{x}_{in}; \mathbf{W}_{i,j}, \mathbf{b}_{i,j})) \quad (10)$$

$$\mathbf{x}_{i,j+1} = \text{MaxPool}(\mathbf{x}_{i,j}) \quad (11)$$

$$\mathbf{x}_{i,j+1} = \mathbf{x}_{i,j+1} + \mathbf{s}_k \quad (12)$$

If $1 < j \leq 5$:

$$\mathbf{x}_{i,j} = \text{ReLU}(\text{Conv}(\mathbf{x}_{i,j-1}; \mathbf{W}_{i,j}, \mathbf{b}_{i,j})) \quad (13)$$

$$\mathbf{x}_{i,j} = \mathbf{x}_{i,j} \oplus \mathbf{x}_{i-1,j} \quad (14)$$

$$\mathbf{x}_{i,j+1} = \text{MaxPool}(\mathbf{x}_{i,j}) \quad (15)$$

$$\mathbf{x}_{i,j+1} = \mathbf{x}_{i,j+1} + s_k \quad (16)$$

$$\begin{cases} j = j + 2 \\ k = k + 1 \end{cases} \text{ return (13), } \quad 0 < j < 5 : \quad (17)$$

$$\mathbf{x}_{i,\text{out}} = \mathbf{x}_{i,j+1}, \quad j = 5$$

In Equations (6)-(16), $\text{Conv}(\cdot)$ represents a convolutional layer, $\text{ReLU}(\cdot)$ represents a Rectified Linear Unit (ReLU) function [40], $\text{Maxpool}(\cdot)$ represents Maxpool layer. $\mathbf{W}_{i,j}$ and $\mathbf{b}_{i,j}$ are weight and bias parameters of corresponding convolutional layers in the j th layer of the i th branch, respectively. Furthermore, $+$ is an addition operation, and $\oplus$ is a concatenation operation.

With the well-designed network structure, MBN can perform multi-scale feature extraction more accurately with the guidance of spectral information. The global spatial distribution is taken into account while maintaining local detail information. Meanwhile, the information of objects at diverse scales can be sufficiently collected, which greatly improves the representation ability of multi-scale features. Additionally, the features obtained also contain spectral signatures, so the connection between these two types of features is tightly strengthened.

D. Optimization

Due to the effects of objects' variability, surrounding environmental characteristics, and sensor errors, land-covers in the same category may have different spectral properties, while those in different classes may have extremely similar spectral properties. As a consequence, it is inaccurate to only apply spectral information for classification. Besides, using only spatial features ignores a large number of spectral features in the image. Misclassification of small-scale targets and over-smoothing phenomenon always occur under such circumstances. Therefore, the combination of spectral features and spatial features is of crucial importance to improve the effect of HSI classification. Most previous studies tended to extract spectral and spatial features separately [41], and in this section, we integrate these two processes into a unified framework. Also, to avoid spectral features being in a weak position, we set the weights of spectral, spatial, and spectral-spatial joint loss function to be the same to balance their contributions to model output.

The architecture of the proposed OSICN is shown in Fig. 1. After getting the output features of spectral information compensation, we utilize ReLU function to optimize the nonlinear representation of features. Additionally, the above spectral-spatial features are flowed into a softmax layer to yield classification maps. Except for the joint main loss function, spectral auxiliary loss function and spatial auxiliary loss function are supplemented for better training. Notice that two auxiliary loss functions and joint main loss function have the same weights to balance the contribution of spectral and spatial information.

Algorithm 1 OSICN

Input:

- 1) An HSI X with ground-truth Y .
- 2) The number of maintained principal components (PCs) p and the size of patches w .

Step 1: For each pixel in the HSI, divide hyperspectral vectors as spectral features.

Step 2: Employ PCA to the HSI and reserve the first p PCs.

Step 3: Around each pixel in the dimension-reduced HSI, extract a patch with the size of $w \times w \times p$ as the spatial feature.

Step 4: Initialize the weights in the network.

Step 5: Input training samples into the network and optimize it with the ADAM optimizer. The ADAM optimizer is utilized with a learning rate of 0.0001, beta_1 of 0.9, and beta_2 of 0.999.

Step 6: For each pixel in the HSI, input the corresponding spectral and spatial features to the network in order to accomplish the classification of the whole image.

Output: A matrix that predicted labels of input HSI.

The complete loss function of the proposed method is defined as follows:

$$\begin{aligned} L = -\frac{1}{M} \sum_{i=1}^M & ([y_i \log(\hat{y}_i^{\text{joint}}) + (1 - y_i) \log(1 - \hat{y}_i^{\text{joint}})] \\ & + [y_i \log(\hat{y}_i^{\text{spectral}}) + (1 - y_i) \log(1 - \hat{y}_i^{\text{spectral}})] \\ & + [y_i \log(\hat{y}_i^{\text{spatial}}) + (1 - y_i) \log(1 - \hat{y}_i^{\text{spatial}})]) \end{aligned} \quad (18)$$

where L is loss function of the proposed OSICN. $\hat{y}_i^{\text{joint}}$, $\hat{y}_i^{\text{spectral}}$, and $\hat{y}_i^{\text{spatial}}$ are predicted labels for the i th training sample, y_i is the true label, and M is the size of the training set. The value of the loss to be minimized will be the weighted sum of all individual losses. The overall network is trained in an end-to-end pattern, whereby all parameters are optimized by an adaptive moment estimation algorithm (ADAM) [42] with a learning rate of $1e-4$ simultaneously. The implementation details of the proposed OSICN are outlined in Algorithm 1.

III. EXPERIMENTS AND ANALYSIS

In this section, we will first describe three public datasets used in this study, after which experimental results and analysis are reported in detail.

A. Dataset Description and Experimental Setting

To show the availability of the proposed method, we conduct classification experiments on three well-known hyperspectral datasets.¹ The first dataset is Pavia University, which consists of 103 spectral bands with 610×340 pixels. The second dataset is Kennedy Space Center (KSC), which consists of 176 bands with 512×614 pixels. The third dataset is Botswana, which consists of 145 spectral bands with 1476×256 pixels. The pseudo-color composition images of three datasets and the corresponding ground-truth maps are shown

¹http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

TABLE I
NUMBER OF TRAINING AND TEST SAMPLES IN THE KSC DATASET

Class number	Class name	Training	Test
1	Scrub	23	738
2	Willow swamp	13	230
3	CP hammock	14	242
4	CP/Oak	14	238
5	Slash pine	5	156
6	Oak/Broadleaf	12	217
7	Hardwood swamp	9	96
8	Graminoid marsh	28	403
9	Spartina marsh	41	479
10	Catail marsh	29	375
11	Salt marsh	31	388
12	Mud flats	39	464
13	Water	81	846
Total		339	4872

TABLE II
NUMBER OF TRAINING AND TEST SAMPLES IN THE BOTSWANA DATASET

Class number	Class name	Training	Test
1	Water	27	243
2	Hippo grass	10	91
3	Floodplain grasses1	25	226
4	Floodplain grasses2	22	193
5	Reeds1	27	242
6	Riparian	27	242
7	Firescar2	26	233
8	Island interior	20	183
9	Acacia woodlands	31	283
10	Acacia shrublands	25	223
11	Acacia grasslands	30	275
12	Short mopane	18	163
13	Mixed mopane	27	241
14	Exposed soils	15	80
Total		330	2918

in Fig. 8 to Fig. 10. For each class in Pavia University dataset, we randomly select 50 samples from ground-truth data to make up the training set, while the remaining samples are employed as the test set. The training set and test set of KSC dataset and Botswana dataset are listed in Table I to Table II.

The experiments are performed repeatedly with the training and test data, and both the average value and standard deviation are reported. Above all, the random training set from the reference data is generated in each repetition. Then, the test set is made up of the remaining reference samples. The performance of each algorithm is evaluated in terms of several widely used measurements, such as Overall Accuracy (OA), Kappa coefficient, training time, and test time. The equations for OA and Kappa coefficient are as follows:

$$OA = \frac{\sum_i x_{ii}}{N} \quad (19)$$

$$Kappa = \frac{N \sum_i x_{ii} - \sum_i (x_{i+} \cdot x_{+i})}{N^2 - \sum_i (x_{i+} \cdot x_{+i})} \quad (20)$$

where x_{ii} denotes the number of pixels correctly assigned to the i th class, x_{i+} and x_{+i} are the number of pixels distributed to the i th class and the true number of pixels of the i th class, respectively, N represents the total number of pixels in HSI, and r is the total category. In order to optimize the network, the ADAM algorithm is employed. The number of training epochs is set to 200, with a learning rate of 0.0001 and batch size

of 128. The experiments are implemented using an NVIDIA GeForce RTX 2080 Ti graphic card.

B. Influence of the Number of Layers in MBN

As the number of layers of the network increases, the expressiveness and abstraction of the network are also improved. However, it is not true that the deeper the network is, the better the classification performance is. Deepening the neural network will bring some negative problems: vanishing gradient, exploding gradient, and over-fitting. Accordingly, the appropriate number of layers is important. In this subsection, we assess the influences of the number of layers in MBN for HSI classification task.

As can be seen from Fig. 4, OA does not always increase with the number of layers in the proposed MBN, but varies in the form of a wave. Fig. 4(a) shows the accuracy of the number of layers in MBN on the Pavia University dataset. When the number of layers is increased from three to four, the accuracy is improved because the obtained feature maps gradually contain more high-level semantic information, which is helpful for classification. However, as the number of layers continues to increase from four, the accuracy decreases instead. This is because directly increasing the number of layers will easily cause a large amount of information loss, resulting in poor prediction results. Therefore, the optimal number of layers in MBN is four for Pavia University dataset.

Fig. 4(b) illustrates the achieved performance in terms of OA and Kappa coefficient for the KSC dataset. It can be found that OA and Kappa coefficient vary relatively smoothly with the number of layers in MBN. The highest accuracy is achieved when the number of layers is six. With less than or more than six layers, the classification effectiveness will be declined due to inadequate feature extraction or too deep networks.

Fig. 4(c) articulates how OA and kappa coefficient change as the proposed MBN deepens for Botswana dataset. The whole trend is more volatile. When the number of layers is five, the accuracy drops to the bottom. Oppositely, when the number of layers is four, the accuracy reaches the maximum. At this point, either increasing or decreasing the number of layers in the MBN leads to an increase in error, i.e., a degradation in network performance and learning ability.

In summary, a reasonable assignment of the number of layers in MBN plays an essential role in raising classification accuracy. This study elaborately designs parameters of network layers for different data sets. Thereby, the feature extraction process can be done in a distinct way, and better classification results can be obtained.

C. Impact of Spectral Information Compensation

HSI is rich in spectral information, so how to make rational use of spectral information is an important issue worth considering. To the best of our knowledge, we are the first to propose an online spectral information compensation method. This method supplements spectral signatures in advance while extracting spatial features, maximizing the use of spectral information. In this subsection, to confirm the effectiveness of

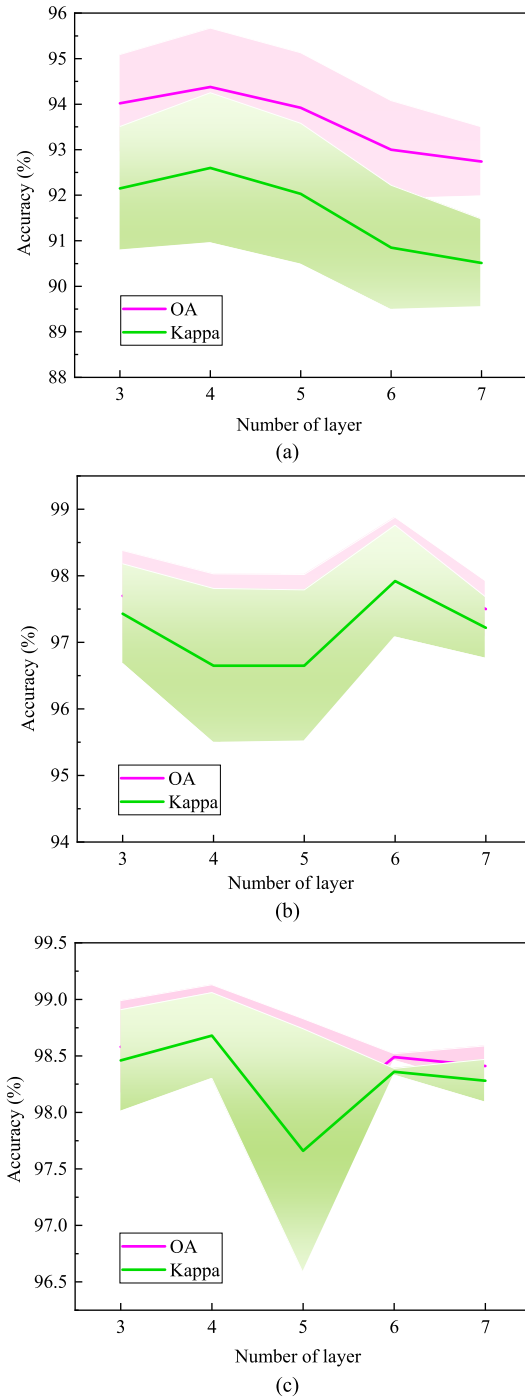


Fig. 4. Accuracy with the number of layer in the proposed MBN on the three classical datasets. (a) Pavia University dataset. (b) KSC dataset. (c) Botswana dataset.

spectral compensation, we perform ablation experiments with and without spectral information compensation.

From Fig. 5, it can be observed that OA with spectral information compensation process is higher than that without it on all three data sets. There is a substantial improvement in OA because spectral information guides the extraction process of spatial information, which makes the acquired feature maps contain the joint spectral-spatial features earlier. In particular, on the Botswana data set, the proposed spectral information compensation method achieves significant performance

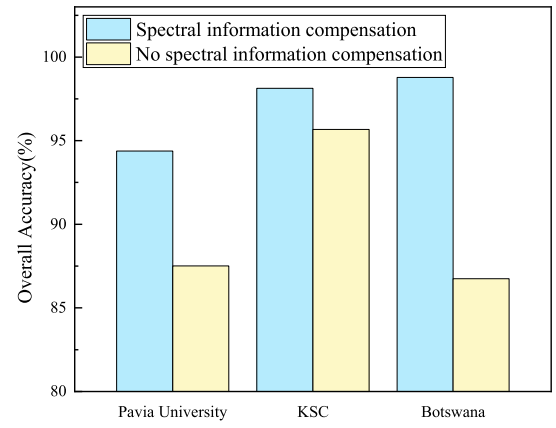


Fig. 5. OA with and without spectral information compensation on three datasets.

improvements of 12.04% in OA. This phenomenon may be attributed to the low inter-class dissimilarity on the Botswana dataset, such as floodplain grasses 1 and floodplain grasses 2, acacia woodlands, acacia shrublands, and acacia grasslands. In this case, spatial information is not enough for precise classification. Thus, compensating spectral information can dramatically increase feature separation and reduce misclassifications, remarkably improving the representation capability and classification performance of the model. The similar phenomenon is also found on the Pavia University dataset. It can be noted that the accuracy improvement of the KSC dataset is minimal in Fig. 5. This is because the inter-class dissimilarity of KSC dataset is higher, and discriminative features can be extracted even if only spatial information is available. Whereas, small-scale land-covers and fine-grained edges are extremely difficult to identify due to the lack of spectral signatures, resulting in the limited classification effect. Hence, OA is further improved when there is spectral information compensation on the KSC dataset.

Based on the above experiments, we can conclude that the spectral information compensation method is beneficial for fine HSI classification.

D. Effects of Different Compensation Strategies

It has been demonstrated that the spectral information compensation method is necessary in subsection C, the next question is where to compensate spectral information to the MBN network. In this subsection, an answer will be given to this question.

Specifically, we design three compensation strategies. Among them, compensation strategy 1 refers to compensating spectral features after the third pooling layer only, compensation strategy 2 means compensating spectral features after the second and third pooling layers, and compensation strategy 3 points to compensating spectral features after all these three pooling layers. The OA of three compensation strategies on the three datasets is shown in Fig. 6. It can be observed that compensation strategy 1 is always superior to compensation strategy 2, and compensation strategy 2 is always better than strategy 3. The reason for this phenomenon exists in the following three aspects: 1) Noise interference.

Spectral features can easily bring about the problem of the phenomenon of “synonyms spectrum” or “foreign objects with the same spectrum”, especially in the detailed feature extraction process. If the spectral information is added at the beginning, the wrong information will keep passing and accumulating in the network, causing accuracy degradation. 2) Inadequate low-level feature extraction. Since spectral information is involved in network learning in advance, low-level spatial feature extraction is influenced by spectral information. That means, the original shallow layers of the network are concerned with extracting low-level spatial details, but now they favor extracting spatial features after the interference of spectral signatures. As a result, spatial detail extraction is incomplete, losing a large number of spatial domain details. 3) Inaccurate high-level semantic modeling. Due to redundant spectral information, the learning process of the network is biased, resulting in incorrect high-level semantics and poor classification effect.

In addition, spectral information is advantageous in guiding high-level features extracted by MBN. Compensating the spectral information only at the last layer is able to directly guide the extraction of high-level semantic features without affecting the learning of the low-level detailed information. The advanced semantic information can obtain information of broader neighborhoods, and with the aid of refined spectral information, the network can contain global features without losing detailed information. Even if there is a deviation in the spectral features at this time, the advanced features extracted by MBN are able to reduce the loss caused by the deviation. The reason is that the high-level features extracted by deeper layers are more robust and have a stronger positive orientation for classification results, making the network have better decision-making ability. In summary, compensation strategy 1 that only compensates spectral information after the last pooling layer can attain the best classification performance.

Actually, we also find an interesting phenomenon: the more spectral information added to the bottom layer, the stronger the interference effect. That is to say, pepper noise is more likely to appear in the lower-level information extraction stage. Compensating spectral information at a deeper level of the network can provide strong support for avoiding the over-smoothing phenomenon by introducing more detailed features. Therefore, it again affirms that compensating spectral information during advanced semantic information extraction is a better choice, which can not only mitigate salt and pepper noise but also ameliorate the over-smoothing phenomenon.

E. Sensitivity of Patch Size and Principal Components

The input patch size and the number of principal components (pcs) affect network classification ability. In detail, a larger input patch size includes more localized information. Nevertheless, when the input patch size is too large, the more interference pixels are contained, the worse performance of feature extraction, which eventually leads to the over-smoothing phenomenon and classification reduction. Similarly, more pcs have richer band information, but too many pcs bring Hughes phenomenon and redundant

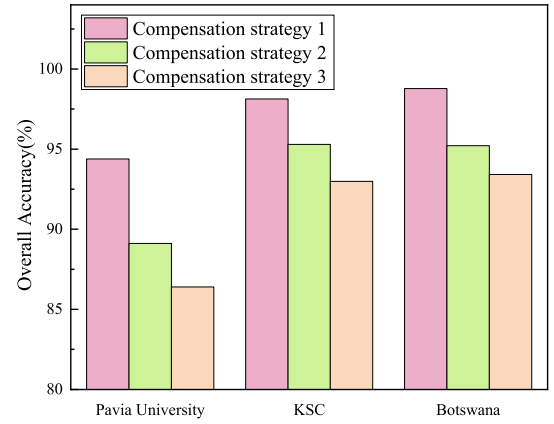


Fig. 6. OA of different compensation strategies in three classical datasets.

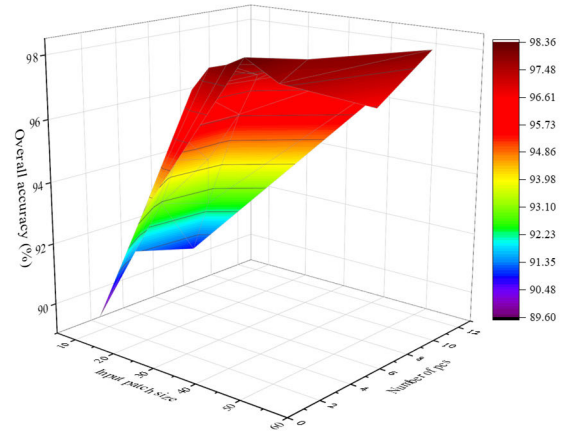


Fig. 7. Performance with different input patch sizes and the number of pcs (with KSC dataset as an example).

information, causing a decrease in accuracy. In this subsection, we analyze the sensitivity of input patch size and the number of pcs for KSC data set as an example.

Fig. 7 verifies the variation pattern of input patch size. It can be seen that as input patch size increases, more locally spatial features are extracted and initial OA dramatically increases; and then gradually decreases due to the introduction of a large number of interfering pixels by the overlarge input patch size. The effect of the number of pcs in terms of OA can also be acquired in Fig. 7. As expected, OA increases overall because the first several components usually include extensive information; after that slightly declines with the increase of pcs as a result of redundant spectral information and more learning parameters.

F. Comparison With the State-of-the-Art Methods

The progressive spectral information compensation approach allows a strong combination of spectral-spatial features. In this subsection, we compare the proposed MBN and OSICN with other methods in limited training pixels: principal component analysis (PCA) [43], Raw [44], extended morphology profile (EMP) [45], 2D CNN [46], spectral-spatial fully convolutional network (SSFCN) [47] and multi-scale dense network (MSDN) [48].

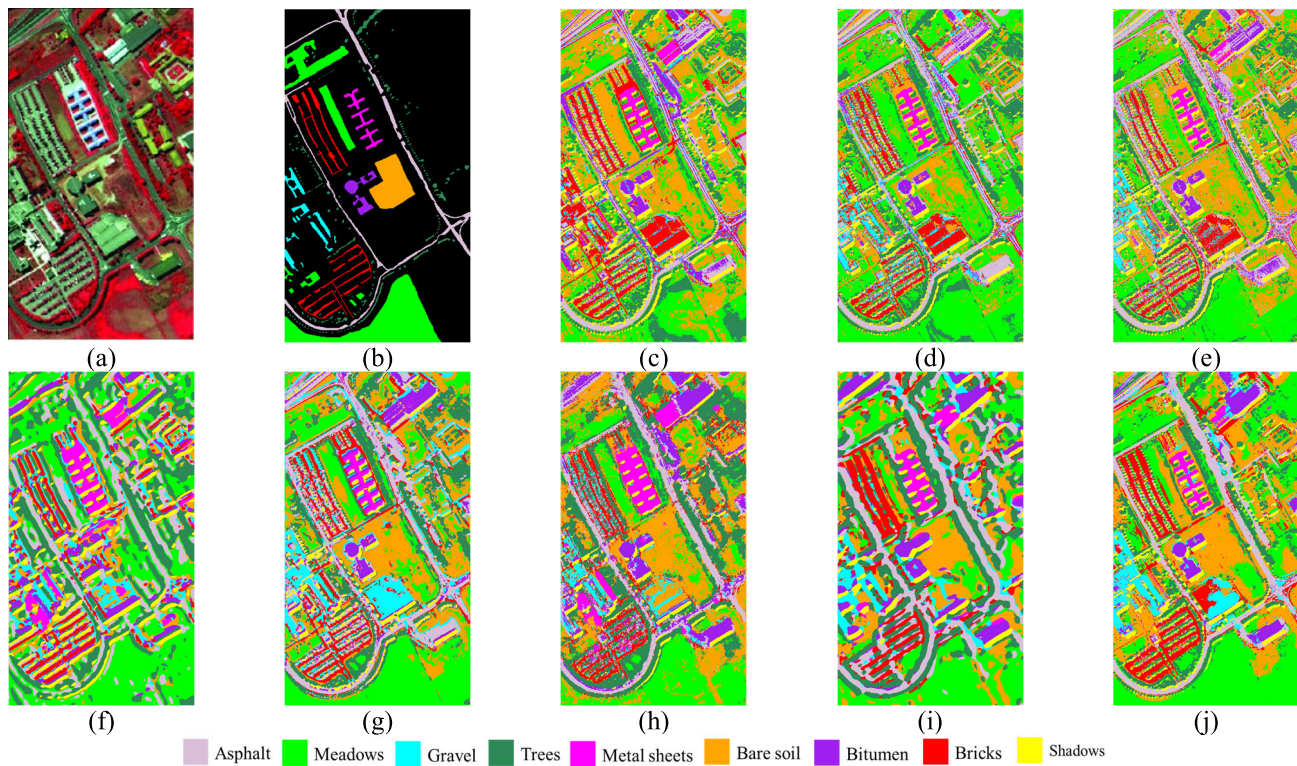


Fig. 8. Classification maps for the Pavia University dataset. (a) Pseudo-color image. (b) Ground-truth map. (c) PCA. (d) Raw (e) EMP. (f) 2D CNN. (g) SSFCN. (h) MSDN. (i) MBN. (j) Proposed.

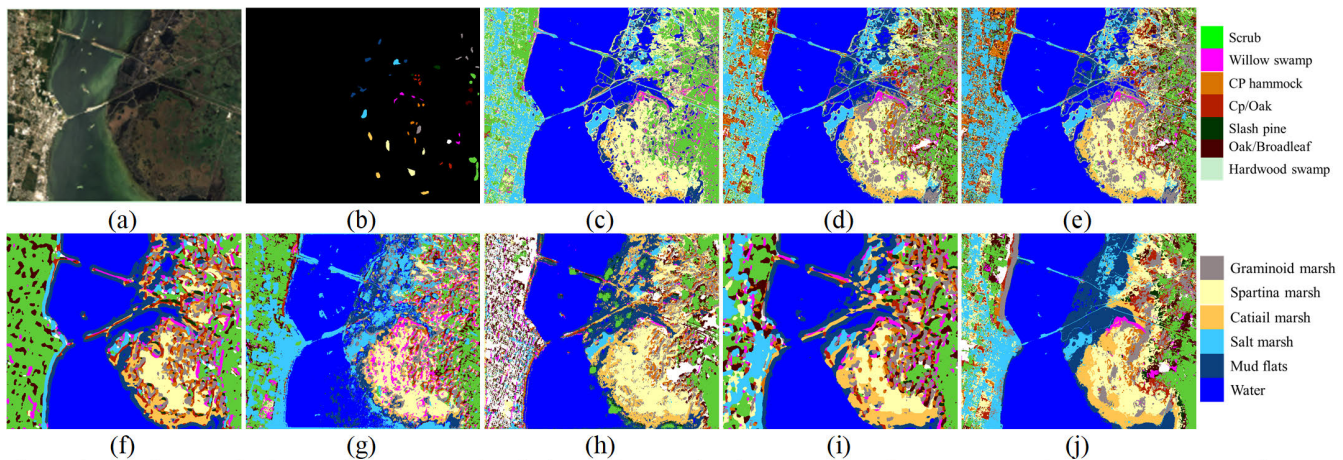


Fig. 9. Classification maps for the KSC dataset. (a) Pseudo-color image. (b) Ground-truth map. (c) PCA. (d) Raw (e) EMP. (f) 2D CNN. (g) SSFCN. (h) MSDN. (i) MBN. (j) Proposed.

Classification maps can visualize the performance of any approach. Figs. 8-10 display the generated classification maps on Pavia University, KSC, and Botswana datasets, respectively. It can be observed that the maps of PCA and Raw always suffer from salt and pepper noise, especially in Pavia University and KSC data sets. Because the above methods lack spatial distribution information, it is difficult to maintain the continuity of objects, leading to a large number of misclassifications. In contrast, the classification maps of spatial classification methods like EMP, 2D CNN, and the proposed MBN are with less noise. This is because these methods take neighboring pixels into account, improving the homogeneity of the

land-covers. However, these classification maps contain a serious over-smoothing phenomenon, which also leads to poor accuracy. Consequently, it is an intuitive idea to combine spectral information and spatial information for HSI classification. There are two typical works of most existing spectral-spatial joint classification methods, e.g. SSFCN and MSDN. Among them, SSFCN represents a category of methods for designing two completely branching networks, and MSDN stands for another category of approaches for directly extracting spectral-spatial information using 3D convolution. Whereas, the classification maps obtained by these methods lose a lot of detailed information. Small-scale targets can barely be

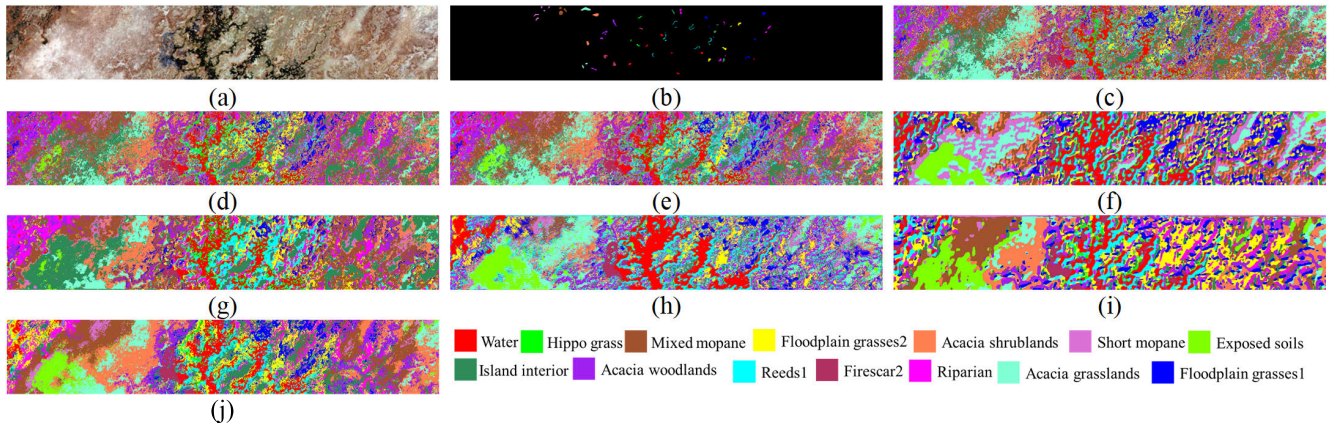


Fig. 10. Classification maps for the Botswana dataset. (a) Pseudo-color image. (b) Ground-truth map. (c) PCA. (d) Raw (e) EMP. (f) 2D CNN. (g) SSFCN. (h) MSDN. (i) MBN. (j) Proposed.

TABLE III
CLASSIFICATION RESULTS OF THE PAVIA UNIVERSITY DATASET

Class	PCA	RAW	EMP	2-D CNN	SSFCN	MSDN	MBN	Proposed
1	61.66±3.27	65.61±2.98	80.50±3.28	64.13±3.78	84.81±2.39	95.21±1.57	85.10±4.13	89.93±2.05
2	49.89±16.60	76.79±5.10	85.47±3.62	73.89±13.63	90.92±3.50	94.06±1.90	84.06±3.36	94.13±3.12
3	19.94±4.73	74.61±5.82	81.51±2.22	57.10±6.77	88.28±3.20	64.83±13.50	79.23±4.75	93.88±3.85
4	94.17±2.42	93.80±2.87	93.42±2.60	86.46±6.40	96.61±1.24	95.38±4.11	95.95±1.94	97.98±1.05
5	99.22±0.16	99.29±0.20	99.21±0.24	99.86±0.20	98.57±1.61	93.51±4.50	98.41±0.87	99.63±0.13
6	62.99±20.58	75.25±6.56	85.46±3.93	35.07±9.84	90.56±3.20	62.90±8.51	93.63±2.86	94.19±3.11
7	92.73±2.08	93.63±1.28	91.86±1.69	66.39±9.67	90.17±4.16	62.34±12.91	93.83±1.82	97.94±1.27
8	83.80±3.22	79.94±8.28	80.77±3.01	84.32±2.26	84.68±6.00	81.11±6.66	90.41±2.76	96.78±1.08
9	99.87±0.05	99.85±0.12	99.84±0.10	96.35±1.59	99.23±0.87	91.49±10.50	96.96±0.93	99.73±0.29
OA (%)	61.74±5.11	77.93±1.63	85.59±1.75	69.83±5.44	90.05±2.45	84.61±3.31	87.51±2.22	94.38±1.29
Kappa×100	53.49±4.51	71.78±1.84	81.28±2.13	61.27±5.73	87.02±3.12	79.94±4.12	83.89±2.77	92.60±1.65
Training Time (s)	0.37±0.15	0.05±0.07	0.38±0.02	28.29±7.68	333.27±81.43	353.66±85.61	19.39±0.72	102.26±8.60
Test Time (s)	4.66±0.54	1.75±0.08	3.33±0.23	0.90±0.22	0.10±0.02	4.51±2.20	7.68±0.43	40.48±2.10

distinguished, such as bricks in Fig. 8(g)-(h), salt marsh in Fig. 9(g)-(h), and floodplain grasses 1 in Fig. 10(g)-(h). This can be explained as follows: Firstly, The method of designing two completely unrelated networks like SSFCN completely separates spectral and spatial domains, and the uniformity of the image is destroyed. As a result, the improvement of classification performance is greatly limited. Secondly, a category of models using 3D convolutional layers, represented by MSDN, suppress spectral feature extraction, and spatial feature extraction dominates the whole process. As a consequence, the over-smoothing phenomenon occurs, giving rise to a decrease in accuracy. In difference to the above methods, the proposed OSICN combines spectral and spatial features in the manner of progressive spectral information compensation. Under this circumstance, information in the spectral and spatial domains can be more closely associated in advance to enhance the representation ability of the model for HSI data. As shown in Fig. 8-10(j), the proposed OSICN attenuates salt and pepper noise and avoids the over-smoothing phenomenon of classification maps. In particular, OSICN maintains fine edges, such as water contours and Acacia shrublands boundaries, even in Botswana datasets with extensive small, similar, irregularly shaped objects illustrated in Fig 10. Thus, the proposed approach can yield better performance than the state-of-the-art methods in the maintenance of internal

continuity and boundary information with fewer training samples.

Table III-V tabulates the effect of different methods from a quantitative perspective for all datasets. Meanwhile, the best classification results are represented in bold. It can be found that the quantitative results in the tables are consistent with the analysis above. The results of spectral classification methods like PCA and Raw are much lower because of massive misclassifications. Spatial classification approaches including EMP, 2D CNN, and MBN induce too much blurring boundaries, so the accuracy is heavily limited. SSFCN and MSDN fuse spectral and spatial features, and then they are better than most spatial or spectral methods. However, there is an exception on the KSC dataset. We can observe that MSDN only gets the OA of 81.54%, which is 11.16% lower than 2D CNN and 14.13% lower than the proposed MBN. Even more, the OA of SSFCN is lower than 2D CNN and the proposed MBN by 14.53% and 17.50%, respectively. The difference in KSC dataset can be probably attributed to the following two points: First of all, KSC dataset has very similar textures in most spectral bands. MSDN adopts 3D CNN, which enlarges the redundancy of spectral bands, which is why it is less effective than spatial methods like 2D CNN and the proposed MBN. Next, SSFCN pays more attention to global information, leading to inaccurate detail feature

TABLE IV
CLASSIFICATION RESULTS OF THE KSC DATASET

Class	PCA	RAW	EMP	2-D CNN	SSFCN	MSDN	MBN	Proposed
1	93.78±1.22	93.78±1.22	87.34±3.62	84.60±4.20	87.00±6.10	98.43±2.44	90.84±4.60	97.49±0.20
2	60.65±4.50	60.65±4.50	83.17±7.12	90.48±5.81	69.69±8.91	97.56±2.62	92.02±6.05	98.91±0.65
3	0.00±0.00	0.00±0.00	89.25±3.52	90.25±5.64	51.94±21.40	72.17±8.48	88.16±6.92	97.93±1.23
4	6.05±7.65	6.05±7.65	67.01±8.14	81.31±6.97	48.19±6.51	73.44±7.61	83.94±8.01	95.16±2.31
5	0.00±0.00	0.00±0.00	41.21±9.41	96.96±3.39	10.44±17.57	92.24±7.04	98.68±2.30	86.53±9.61
6	0.09±0.29	0.09±0.29	49.07±8.37	72.97±12.96	30.32±32.63	69.78±6.78	95.57±3.10	91.24±2.76
7	0.00±0.00	0.00±0.00	82.81±7.87	86.31±10.23	16.25±34.28	25.09±12.50	94.71±5.94	100.00±0.00
8	30.19±7.52	30.19±7.52	90.09±2.93	94.50±3.76	75.08±11.64	80.68±5.53	96.16±3.13	98.88±0.86
9	90.87±4.46	90.87±4.46	95.53±1.95	97.83±2.81	95.82±4.46	74.70±6.09	98.14±2.04	100.00±0.00
10	70.24±3.25	70.24±3.25	93.92±2.40	97.36±2.84	74.24±5.77	81.95±5.65	99.19±1.16	99.86±0.13
11	89.22±2.17	89.22±2.17	97.03±1.46	97.13±3.07	98.58±1.31	99.97±0.08	99.08±1.75	98.06±1.93
12	72.11±1.19	72.11±1.19	91.31±3.34	94.97±1.57	87.41±3.55	92.05±1.57	97.62±2.20	99.78±0.21
13	99.96±0.05	99.96±0.05	99.97±0.04	100.00±0.00	99.29±0.77	95.19±1.16	100±0.00	100.00±0.00
OA (%)	65.53±0.39	65.53±0.39	87.86±0.68	92.70±0.92	78.17±2.34	81.54±2.67	95.67±1.20	98.13±0.75
Kappa×100	60.69±0.50	60.69±0.50	86.49±0.75	91.87±1.02	75.59±2.64	79.48±2.93	95.19±1.34	97.92±0.84
Training Time (s)	0.32±0.17	0.32±0.17	0.06±0.04	46.54±28.73	507.74±83.03	188.73±6.86	22.72±2.03	90.55±0.00
Test Time (s)	0.78±0.36	0.78±0.36	0.24±0.01	0.52±0.43	0.12±0.02	232.15±0.57	1.59±0.37	6.94±0.70

TABLE V
CLASSIFICATION RESULTS OF THE BOTSWANA DATASET

Class	PCA	RAW	EMP	2-D CNN	SSFCN	MSDN	MBN	Proposed
1	99.96±0.11	99.90±0.18	99.87±0.45	99.25±1.96	99.17±0.63	78.13±13.88	82.92±33.04	100.00±0.00
2	67.58±11.15	92.89±6.24	96.61±2.26	50.43±8.49	87.21±6.76	79.88±23.07	81.86±22.20	99.08±1.72
3	94.43±2.39	97.54±1.79	98.67±0.78	89.91±5.10	96.90±2.92	93.53±4.61	85.39±28.78	99.88±0.26
4	72.66±28.19	95.33±2.87	94.45±2.67	60.20±10.86	92.36±6.54	87.04±5.98	77.77±37.84	98.14±1.29
5	77.68±2.46	84.26±3.26	84.26±5.02	80.90±4.15	87.86±1.89	83.24±7.77	77.06±25.45	96.76±1.83
6	55.40±9.66	76.28±4.93	76.95±5.18	79.33±11.09	87.11±3.80	79.46±9.00	90.12±12.13	98.55±0.97
7	94.17±2.04	97.02±1.47	97.02±0.96	83.26±7.85	95.94±4.16	91.00±12.52	89.05±19.58	97.42±4.46
8	84.83±4.82	96.09±3.27	97.01±1.57	92.29±4.26	95.28±5.43	89.73±7.78	92.51±5.32	100.00±0.00
9	73.88±3.10	86.92±2.09	88.58±3.54	87.49±4.74	91.16±3.29	83.65±3.93	98.02±2.87	99.05±0.92
10	73.13±3.15	87.06±7.11	87.47±3.85	91.43±5.91	97.63±2.52	96.78±1.67	80.00±40.00	99.36±0.67
11	92.27±2.09	93.11±2.80	93.23±3.75	93.38±8.74	99.47±0.98	97.47±3.70	84.58±30.81	99.03±0.68
12	94.37±1.69	93.86±2.86	93.44±2.38	88.89±4.71	95.76±2.56	89.85±5.26	93.98±12.57	98.92±1.54
13	79.35±2.66	90.39±3.32	90.39±3.13	84.81±6.48	93.47±5.05	96.62±2.19	89.29±29.79	98.13±1.62
14	98.75±0.92	98.07±1.88	97.69±2.33	92.75±7.64	93.86±4.69	95.08±6.82	92.25±8.78	99.47±1.19
OA (%)	82.41±1.56	91.42±0.89	91.84±0.86	85.29±1.38	94.06±1.08	87.03±3.69	86.74±19.44	98.78±0.35
Kappa×100	80.93±1.70	90.70±0.96	91.16±0.93	84.05±1.50	93.56±1.18	85.96±3.98	85.66±20.99	98.68±0.38
Training Time (s)	0.01±0.00	0.01±0.00	0.14±0.13	9.29±0.67	661.81±90.47	10870.96±421.22	16.35±1.01	77.73±8.36
Test Time (s)	0.16±0.00	0.11±0.00	0.13±0.01	0.33±0.55	0.21±0.22	395.88±13.45	1.13±0.26	3.87±0.73

learning, such as incomplete narrow borders of buildings, river contours, etc. Therefore, 2D CNN and the proposed MBN still outperform spectral-spatial method SSFCN. In comparison, the proposed spectral-spatial joint classification method OSICN achieves better performance than the above methods. Through progressive compensation of spectral information, spectral features are able to participate in learning in advance and guide the extraction of spatial features. Hence, spectral and spatial information can be efficiently inherited and integrated to get a prominent improvement in classification accuracy, as observed in the experimental results.

Apart from classification maps and accuracy, runtime time is also an essential factor for HSI classification. From Tables III-V, it can be noticed that the training time and test time for all the datasets of the proposed method are much less than those of MSDN. This phenomenon is particularly evident in the Botswana dataset: the runtime of MSDN is about 11267 seconds, and that of the proposed OSICN is only about 82 seconds. Although the runtime of MSDN is approximately 138 times as much as the proposed method, OA of MSDN is still at least 10% lower than the proposed OSICN. In the case of popular SSFCN, it employs fully convolutional architecture

and does not need to extract patches from HSI, thus having a great efficiency advantage. Even so, the proposed method costs less time than SSFCN. Take the KSC dataset as an example, the runtime of the proposed OSICN is about only 98 seconds, while that of SSFCN is about 508 seconds. While in the Botswana dataset, SSFCN runs even about 7 times longer than the proposed method. Similar trends can be perceived in other datasets. Moreover, we can also find that approaches with the SVM classifier (i.e., PCA, Raw, and EMP) always cost less runtime than deep learning-based methods, which shows the excellence of traditional classifiers from the perspective of time consumption. Nevertheless, the classification effect of these methods is not satisfactory. In fact, it is also not wise to trade a short time for a steep decline in accuracy. On the contrary, the proposed OSICN shows a better balance among classification maps, accuracy, and time cost.

IV. CONCLUSION

In this article, we propose an online spectral information compensation network with a pixel-to-pixel, end-to-end architecture for HSI classification. To begin with, we propose

the concept of candidate spectral vectors to serve as support for online compensation. Moreover, a novel approach, e.g. progressive spectral information compensation, is proposed for tackling the problem of inadequate fusion between spectral and spatial information. Spectral signatures can be fed into the network in an earlier stage to participate in feature learning, and assist in the extraction of spatial features to lessen over-smoothing and misclassification. In addition, we design MBN with multiple branches and multi-level extraction. With this structure, the complete internal continuity of larger-scale land-covers and clear boundaries of smaller-scale targets can be guaranteed at the same time. In the above process, we discover a fascinating phenomenon: the direct combination of spectral and spatial information will instead bring interference and undermine classification performance. Therefore, the way of spectral-spatial combination needs more exploration, and the proposed model provides an idea for inline compensation of spectral signatures. Eventually, two auxiliary loss functions and joint main loss function are designed to alleviate the imbalance of spectral and spatial information extraction. Experimental results on three benchmark HSI datasets with limited training samples also prove the superiority of the proposed approach.

Our future work will try to introduce transfer learning with deep neural networks to reinforce spatial feature extraction. Since the spatial resolution of HSI is much lower compared to the spectral resolution, the mining of spatial information is more difficult. Thus, how to boost both spectral and spatial feature extraction is worthy of further study.

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